

SMART BLIND STICK USING ULTRA SONAR SENSORS WITH VOICE ASSISTANCE

A PROJECT REPORT

SUBMITTED BY

MAMIDI SAI VENKATA VIGNESH (21BAS1049)

SAMANDEEP KOUR (21BAS1025)

RAMKUMAR K S (21BAS1015)

MOHAN SAI REDDY (21BAS1050)

HEMNATH A (21BAS1017)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

AEROSPACE ENGINEERING



**CHANDIGARH
UNIVERSITY**

Discover. Learn. Empower.

CHANDIGARH UNIVERSITY

MAY 2022



BONAFIDE CERTIFICATE

Certified that this project report “ **SMART BLIND STICK USING ULTRA SONAR SENSORS WITH VOICE ASSISTANCE** ” is the bonafied of **Mamidi Sai Venkata Vignesh, Samandeep Kour, Ramkumar K S, Hemnath A, Mohan Sai Reddy** who carried out the project work under my supervision.

SIGNATURE

Dr. Rajneesh Kumar Kaushal
SUPERVISOR
Assistant Professor
Academic Unit 3

SIGNATURE

Dr. Gyanendra Singh Goindi
HEAD OF THE DEPARTMENT
Academic Unit 3

Submitted for the project viva-voice examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

It is our immense pleasure to express our sincere gratitude to our esteemed mentor and supervisor **Dr. RAJNEESH KUMAR KAUSHAL** whose guidance, encouragement, suggestion, cooperation and patience throughout the course of work helped us in the accomplishment of this project. We offer our humble thanks and shall always be obliged to him.

We are also grateful to **Dr. Gyanendra Singh Goindi**, Head Of Department of Academic Unit 3, Chandigarh University, Gharuan for timely providing us all the facilities and motivating us in the completion of experimental work. We owe thanks to the management of Chandigarh University for providing us the required infrastructure.

At last ,we just say that if we commit ourselves to the successful completion of a task, then we personify excellence and we are extremely grateful to our parents and friends for supporting us and for their love they provided us during the whole project.

TABLE OF CONTENTS

CONTENTS		Page No.
1	Acknowledgement	1
2	Table of contents	2-3
3	List of figures	4
4	Abstract	5
	Chapter -1 Introduction	
1.1	Introduction to Smart Blind Stick	6-7
1.2	Relevant Contemporary Issues	7
1.3	Task Identification and Problem Identification	7-8
	Chapter -2 Literature Survey	
2.1	Ultrasonic Walking Stick for Visually Impaired People Manisha Gend1 , Sana Pathan2 , Charushila Shedge3 , Prof. D. P Potdar4, Volume 6, Issue 1, June 2021	9
2.2	Smart Ultrasonic Walking Stick for Visually Impaired People Manisha Gend1 , Sana Pathan2 , Charushila Shedge3 , Prof. D. P Potdar	9
2.3	Smart Blind Stick Design and ImplementationAmira. A. Elsonbaty	9-10
2.4	smart blind stick using ultra sonic sensor ,Talari tirupal ,August 2021	10-11
2.5	smart blind stick using ultra sonic sensors ,25-Jun-2018	11
2.6	An IOT based voicecontrolled blind stick to guide blind people ,Abhijit Pathak , volume 9, jan 2020	11-12
2.7	Aritra Ray and Hena Ray (2019)	12-13

	Chapter -3 Design Flow/ Process	
3.1	Features/Specifications	14
3.2	Design constraints	14-19
3.3	Design flow/Flowchart	20

	Chapter -4 Result Analysis And Validations	
4.1	Methodology	21
4.2	End result and the output	22
	Chapter -5 Conclusion and Future Work	
5.1	Conclusion	23
5.2	Future scope	23-24
6	References	25-26

LIST OF FIGURES

S.no	Title
Fig. 1	Circuit diagram
Fig. 2	Ultra Sonic Sensors
Fig. 3	Breadboard
Fig. 4	Jump Wires
Fig. 5	Flowchart
Fig. 6	Blind Stick with Ultra Sonic Sensors.

ABSTRACT

The main aim of this paper is to assist blind persons without human need. Notably, the visually impaired individuals convey a hand that stays with them at whatever point they need help. Once in a while in any event, when they utilize this stick, there is no assurance that the visually impaired people are protected and get in arriving at their destinations. There might be a deterrent in their way yet isn't experienced by the individual with the assistance of the stick. Notably, the visually impaired individuals convey a hand that stays with them at whatever point they need help. Once in a while in any event, when they utilize this stick, there is no assurance that the visually impaired people are protected.

There might be an obstruction in their way however isn't experienced by the individual with the assistance of the stick. Thus, the people may be injured if the obstacle is big enough or dangerous. Thus, in this paper, a blind stick is designed and developed to assist the blind person and provide them a clear path. The system consists of an ultrasonic sensor fixed to the user's stick.

While the user moves the stick in the forward direction, the ultrasonic sensor with Arduino Nano fixed to the stick tries to detect the obstacle if any present in the path. If the sensor recognizes the obstacle, the output of the recipient triggers, and this change will be identified by the microcontroller since the output of the receiver is given as inputs to the microcontroller. This stick recognizes the article before the individual and offers a reaction to the client either by vibrating or through the order. In this way, the individual can walk with no fear. This gadget will be the best answer for defeat the troubles of the visually impaired individual.

CHAPTER -1

INTRODUCTION

1.1. Introduction to Smart Blind Stick

According to the WHO, 30 million socioeconomic classes are permanently impaired, and 285 billion people have vision problems. If you notice them, you can assume that they are unable to walk without assistance. One must request that direction be delivered to their desired location. They must face more challenges in their daily lives. The horizontal range of the visual field with both eyes open is less than or equal to 20 degrees for those with visual acuteness of 6/60. These people are classified as blind.

Visually impaired people struggle to engage with and sense their surroundings. They have little interaction with their environment. Physical mobility might be difficult for visually impaired people since it can be difficult to discern between objects.

Researchers have spent decades developing a clever and smart stick to help visually impaired people avoid obstacles and provide location information.

An individual who uses this visually challenged stick can walk more confidently. This strolling stick is an alternative to the traditional strolling stick. Over the last few decades, scientists have been working on new gadgets to create a good and dependable system for visually impaired people to identify obstacles and notify them when they are in danger.

A smart walking stick is meant to detect obstructions, allowing the blind to traverse safely. The auditory messages will keep the user awake and prevent accidents significantly. A voice-activated automated switching system is also included to assist them in their own area. This technology demonstrates an idea for providing smart technological assistance to blind persons in both public and

private spaces. The ultrasonic sensor, Arduino nano, breadboard, jump wires, and speaker are all included in the proposed system.

When ultrasonic waves collide with an object and the signal is returned to the sensor, the suggested system detects the impediment. The ultrasonic sensor in the Stick measures the distance between objects and the smart walking stick.

When any foreign objects or impediments come within range of an ultrasonic sensor, the buzzer begins to beep, alerting the person to the presence of the obstacle. The smart walking stick is a straightforward and straightforward device.

It gives the person the best travel assistance. The blind individual can go from one location to another without the assistance of others. The system's primary goal is to develop an effective navigation tool for blind people that provides a sensation of vision by providing information about their environment and objects.

1.2. Relevant Contemporary Issues And Problem Identification

The previous blind stick cannot help the person to know about the coming obstacles and this restricts their movements. Due to this a blind person has to depend on others for every single movement.

They depend on their families for mobility and financial support. Their mobility opposes them from interacting with people and social activities. In the past, different systems are designed with limitations without a solid understanding of the no visual perception.

The threat of being attacked or fell down and injure oneself increases.

1.3. Task identification

The main aim of our project is to make a smart blind stick with voice based system by which it can warn the person from a distance of atleast 10 to 12 inches if some obstacles come in their way by the sound of a beep.

The following are the main objectives of our project which will help us to achieve our goal:

- To design the circuit diagram, coding algorithm and model designing. It will help us to get an idea that how our project is going to look and work.
- Testing each part separately and ensuring that everything works perfectly before reaching the final point of assembling all the parts.
- Assembling all the parts together and checking all the connections and demonstrating the final project and its working

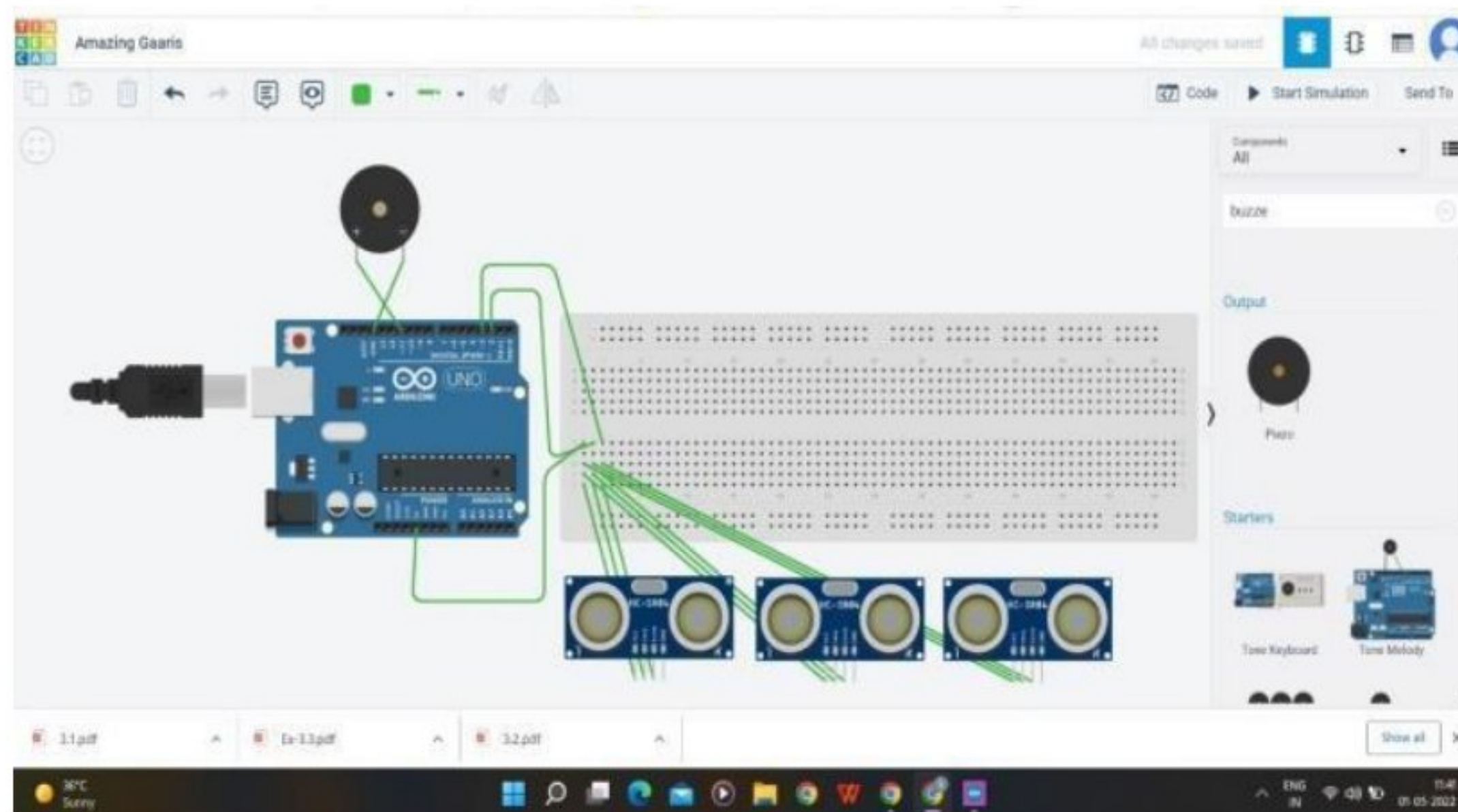


Fig (1) Circuit Diagram

CHAPTER -2

LITERATURE SURVEY

2.1. Ultrasonic Walking Stick for Visually Impaired People Manisha Gend1 , Sana Pathan2 , Charushila Shedge3 , Prof. D. P Potdar4, Volume 6, Issue 1, June 2021

The study focus on a simple method of detecting the obstacle and route by using an ultrasonic sensor that can detect a hole or stair with maximum range about 2 meter. As we can see Blind people is having their trouble to do their life routines because they can't see even a single things. With our idea, we want to help this kind of people to live their life freely. This ultrasonic blind stick have a several feature that surely can help this blind people to navigate routes and detect an obstacle that surely can make their life routines easier. The user just need to use the blind the normal blind stick , the different is , blind people can detect a hole or stair more faster and easily. Besides that, guardian or user of the blind waking stick can find their stick by remote if they misplaced them Smart.

2.2 Smart Ultrasonic Walking Stick for Visually Impaired People Manisha Gend1 , Sana Pathan2 , Charushila Shedge3 , Prof. D. P Potdar

Blind people can't easily recognize obstacles or stairs while using normal blind stick. The blind traveler should depend on any other guide like blind cane , people information , trained dogs, etc. About the 90% of the worlds visually impaired live in developing countries. No safety features on the normal blind stick.

2.3. Smart Blind Stick Design and ImplementationAmira. A. Elsonbaty

It is an ultrasonic sensor, also known as an ultrasonic transducer, that is based on a transmitter and receiver and is primarily used to determine the distance from a target object with a wavelength ranging from 20kHz to 20 MHz [5].

Ultrasonic sensors, like sonar detectors, work by transmitting a pulse of sound outside the range of human hearing. At the speed of sound (340 m/s), this pulse travels away from the range finder in a conical shape. The sound bounces off an object and is reflected in the range finder. This is interpreted as an echo by the sensor, which measures the time between transmitting the signal and receiving the echo. The object's distance is then calculated using this interval by a controller in simple notation[6] Ultrasonic Sensors also known as transceivers when they both send and receive work on a principle similar to radar or sonar which evaluate attributes of a target by interpreting the echoes from radio or sound waves respectively. Ultrasonic sensors generate high frequency sound waves and evaluate the echo which is received back by the sensor. [Sensors](#) calculate the time interval between sending the signal and receiving the echo to determine the distance to an object. This technology can be used for measuring: wind speed and direction (anemometer), fullness of a tank and speed through air or water. For measuring speed or direction a device uses multiple detectors and calculates the speed from the relative distances to particulates in the air or water. To measure the amount of liquid in a tank, the sensor measures the distance to the surface of the fluid. Further applications include: humidifiers, sonar, medical ultrasonography, burglar alarms and non-destructive testing. Systems typically use a transducer which generates sound waves in the ultrasonic range, above 18,000 hertz, by turning electrical energy into sound, then upon receiving the echo turn the sound waves into electrical energy which can be measured and

displayed. The technology is limited by the shapes of surfaces and the density or consistency of the material.

2.4. smart blind stick using ultra sonic sensor ,Talari tirupal ,August 2021

A buzzer is a little yet productive segment to add sound highlights to our undertaking/framework. It is minuscule and minimized 2-pin structure henceforth can be effortlessly utilized on a breadboard, Perf Board and surprisingly on PCBs which makes this a broadly utilized segment in most electronic applications

2.5. smart blind stick using ultra sonic sensors ,25-Jun-2018

It is used to produce a clock of precise frequency using constant voltage. There is one limitation of using Arduino Nano i.e. it doesn't come with a DC power jack, which means you can not supply an external power source through a battery. It has 357 kb of storage to save and execute the code from Aurdino IDE Arduino Nano has 8 analogue pins. It has 6 PWM pins among the digital pins. It has a crystal oscillator of 16MHz. It's operating voltage varies from 5V to 12V . The Arduino Nano is a small, complete, and breadboard-friendly [board](#) based on the [ATmega328P](#) released in 2008. It offers the same connectivity and specs of the [Arduino Uno](#) board in a smaller form factor.[\[1\]](#)

The Arduino Nano is equipped with 30 male [I/O](#) headers, in a [DIP-30](#)-like configuration, which can be programmed using the [Arduino](#) Software [integrated development environment](#) (IDE), which is common to all Arduino boards and running both online and offline. The board can be powered through a [type-B mini-USB](#) cable or from a 9 V battery.[\[2\]](#)

In 2019, Arduino released the **Arduino Nano Every**, a pin-equivalent evolution of the Nano. It features a more powerful ATmega4809 processor and twice the

RAM.[\[3\]](#)

2.6. An IOT based voice controlled blind stick to guide blind people ,Abhijit Pathak , volume 9, jan 2020

Arduino is an [open-source hardware](#) and [software](#) company, project, and user community that designs and manufactures [single-board microcontrollers](#) and [microcontroller](#) kits for building digital devices. Its hardware products are licensed under a [CC BY-SA license](#), while software is licensed under the [GNU Lesser General Public License](#) (LGPL) or the [GNU General Public License](#) (GPL).[\[1\]](#) permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially from the official website or through authorized distributors.

Arduino board designs use a variety of [microprocessors](#) and controllers. The boards are equipped with sets of digital and analog [input/output](#) (I/O) pins that may be interfaced to various expansion boards ('shields') or [breadboards](#) (for prototyping) and other circuits. The boards feature serial communications interfaces, including [Universal Serial Bus](#) (USB) on some models, which are also used for loading programs. The microcontrollers can be programmed using the [C](#) and [C++ programming languages](#), using a standard API which is also known as the **Arduino language**, inspired by the [Processing language](#) and used with a modified version of the Processing IDE. In addition to using traditional [compiler toolchains](#), the Arduino project provides an [integrated development environment](#) (IDE) and a command line tool developed in [Go](#).

2.7. Aritra Ray and Hena Ray (2019)

Aritra Ray and Hena Ray (2019) reported that they is no worthy prototype production for visual impaired. The authors developed the MEMS sensor based and low weight walking aid for the scribes benefit. Namita Agarwal et al. (2015) proposed a safe and secure electronic guiding stick for scribes. The navigation system provides the information to the scribes through their headset for both indoor and outdoor conditions. Agrawal and Gupta (2018) developed a user friendly and vibration based smart aid stick for visual impair. The authors concentrated on low cost and mass production system for low economic people point of view. Arora and Gaikwad (2017) provided the theoretical model for smart electronic aid to the scribes. The theoretical model utilized the IR and ultrasonic sensor for hurdle detection and the GPS module for the voice assistance to the desired location and emergency conditions. Divya et al. (2019) described the ultrasonic based walking aid for scribes using Arduino micro-controller. In this work, the GSM module provides the alert message. The alert signal is generated by the Buzzer sound to the scribes. The purpose of GPS module is to give the location to the guardian, when the scribes in emergency or threat. The authors reported that the cost of the system is Rs.

ISSN: 2005-4297 IJCA Copyright © 2020 SERSC International Journal of Control and Automation Vol. 13, No. 2, (2020), pp. 512 - 519 513 2,500 and the battery back of 12 hours. Liang-Bi Chen et al. (2019) presented the paper on wearable smart glass smart walking aid for scribes. In this work, deep learning tool used to intelligent reorganization of front image or any signals. Mohammed Noman et al. presented a smart glass model incorporating with smart phone for scribes. The special features of this work to recognize the English and Arabic language for face reorganization and in-door navigation. The authors reported that their model cost

ranges from 245 USD to 1900 USD. A project reported by Mohammadi et al. (2013) explained the detailed design and implementation of a Self-Energized Smart Vision Stick for Visually Impaired People. The device consists of three main systems which are detection and alert, emergency, and renewable power generation. The detection and alert system are composed of an ultrasonic sensor that detects surrounding obstacles and sends a signal to a microcontroller for processing. The distance of potential obstacles is calculated and feedback is given to the user accordingly in order to avoid the potential obstacle.

Chapter -3

Design Flow/ Process

3.1. Features/Specifications

Following are the specifications/ features of our project which makes it more effective than the usual blind stick:

- This stick is user – friendly and cost efficient.
- Easy to use and handle.
- Battery backup of 20,000 mAH.
- Operated by just one click.
- Can detect the range of upto 200 inches.
- Obstacle detection with 180 degree sensing technology.
- Use of advanced technology so that it will weight less.
- Used by simple codes and easily customizable for different terrain.
- Totally eco-friendly as it does'nt contain any hazardous waves or materials.
- No need of training to use it need to aware that sound indicates the obstacle.
- Capable of detecting moving objects.

3.2. Design constraints

The hardware which are needed for the successful fulfilment of this project are:

- **ARDUINO NANO**
 1. The Arduino Nano can be powered by means of the Mini-B USB connection, 6-20V unregulated exterior strength grant (pin 30), or 5V regulated external power supply (pin27).
 2. The power supply is routinely selected to the best voltage source. The ATmega328 has 32 KB, (also with two KB used for the bootloader. The ATmega328 has two KB.

3. Each of the 14 digital pins on the Nano can be used as an enter or output, using `pinMode()`, `digitalWrite()`, and `digitalRead()` functions. They operate at 5 volts.
4. Each pin can provide or receive a maximum of forty mA and has an inside pull-up resistor (disconnected by using default) of 20-50 kOhms. In addition, some pins have specialized functions:
 - Serial: 0 (RX) and 1 (TX). Used to acquire (RX) and transmit (TX) TTL serial data. These pins are linked to the corresponding pins of the FTDI USB-to-TTL Serial chip.
 - External Interrupts: two and 3 These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a alternate in value. See the `attachInterrupt()` feature for details.
 - PWM: 3, 5, 6, 9, 10, and 11 Provide 8-bit PWM output with the `analogWrite()` function.
 - SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins guide SPI communication, which, although furnished by using the underlying hardware, is not presently included in the Arduino language.
 - LED: thirteen There is a built-in LED linked to digital pin 13 When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

The Nano has 8 analog inputs, each of which supply 10 bits of decision (i.e. 1024 special values). By default they measure from ground to 5 volts, even though is it feasible to change the top quit of their vary using the `analogReference()` function. Analog pins 6 and 7 can't be used as digital pins. Additionally, some pins have specialised functionality:

- I2C: A4 (SDA) and A5 (SCL). Support I2C (TWI) verbal exchange using the Wire library (documentation on the Wiring website).

There are a couple of different pins on the board:

- AREF. Reference voltage for the analog inputs. Used with analogReference().
- Reset. Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

- **PROGRAMMING**

1. The Arduino Nano can be programmed with the Arduino software. Select "Arduino Duemilanove or Nano w/ ATmega328" from the Tools > Board menu (according to the microcontroller on your board).
2. The ATmega328 on the Arduino Nano comes pre burned with a bootloader that permits you to upload new code to it without the use of an exterior hardware programmer
3. it communicates using the unique STK500 protocol. You can additionally bypass the bootloader and software the microcontroller via the ICSP(In Circuit Serial Programming) header the usage of Arduino ISP or similar and it has automatic software program reset.

- **BUZZER**

The buzzer is a sounding device that can convert audio signals into sound signals. It is normally powered with the aid of DC voltage. It is extensively used in alarms, computers, printers and other digital merchandise as sound devices. It is in general divided into piezoelectric buzzer and electromagnetic buzzer, represented through the letter "H" or "HA" in the circuit. According to extraordinary designs and uses, the buzzer can emit various sounds such as music, siren, buzzer, alarm, and electric bell.

- **ULTRA SONIC SENSORS**

An ultrasonic sensor is an instrument that measures the distance to an object the use of ultrasonic sound waves.

An ultrasonic sensor makes use of a transducer to send and get hold of ultrasonic pulses that relay lower back facts about an object's proximity.

High-frequency sound waves mirror from boundaries to produce distinct echo patterns. Below the fig (1) represents the ultra sonic sensors.



Fig (2) Ultra Sonic Sensors

- **BREADBOARD**

Breadboards are one of the most critical portions when mastering how to build circuits. In this tutorial, you will analyze a little bit about what breadboards are, why they are called breadboards, and how to use one. Once you are carried out you ought to have a simple appreciation of how breadboards work and be able to build a basic circuit on a breadboard.

An electronics breadboard (as hostile to the type on which sandwiches are made) is clearly referring to a solderless breadboard. These are remarkable gadgets for making temporary circuits and prototyping, and they require surely no soldering.

Prototyping is the process of trying out out an thought by way of developing a preliminary model from which different varieties are developed or copied, and it is one of the most common makes use of for breadboards. If you aren't positive how a circuit will react beneath a given set of parameters, it's high-quality to construct a prototype and take a look at it out.



Fig (3) Breadboard

- **JUMPER WIRES**

1. A jumper wire is a quick insulated wire with bare (stripped of insulation) ends.
2. You use jumper wires, such as the one shown in the following figure, to connect two factors in a breadboard circuit.
3. Even if you have a set of precut jumper wires, chances are you will want to make a jumper wire of a precise length for a circuit or two.



FIG (4) Jumper Wires.

- **POWER SUPPLY**

1. 20000mAH.
2. USB rechargeable.
3. Can be used as a power source for Arduino nano.
4. Power banks work by means of the use of state-of-the-art electronics to control taking in charge from a charger, storing it in a battery and then charging other devices.
5. Power banks are now not simply a easy battery: they use state-of-the-art electronics circuitry to control being charged, and then charging other devices.

- **ARDUINO IDE**

1. The Arduino Integrated Development Environment - or Arduino Software (IDE) - consists of a textual content editor for writing code, a message area, a text console, a toolbar with buttons for frequent functions and a collection of menus.
2. It connects to the Arduino hardware to add packages and talk with them.

3.3. Design flow/Flowchart

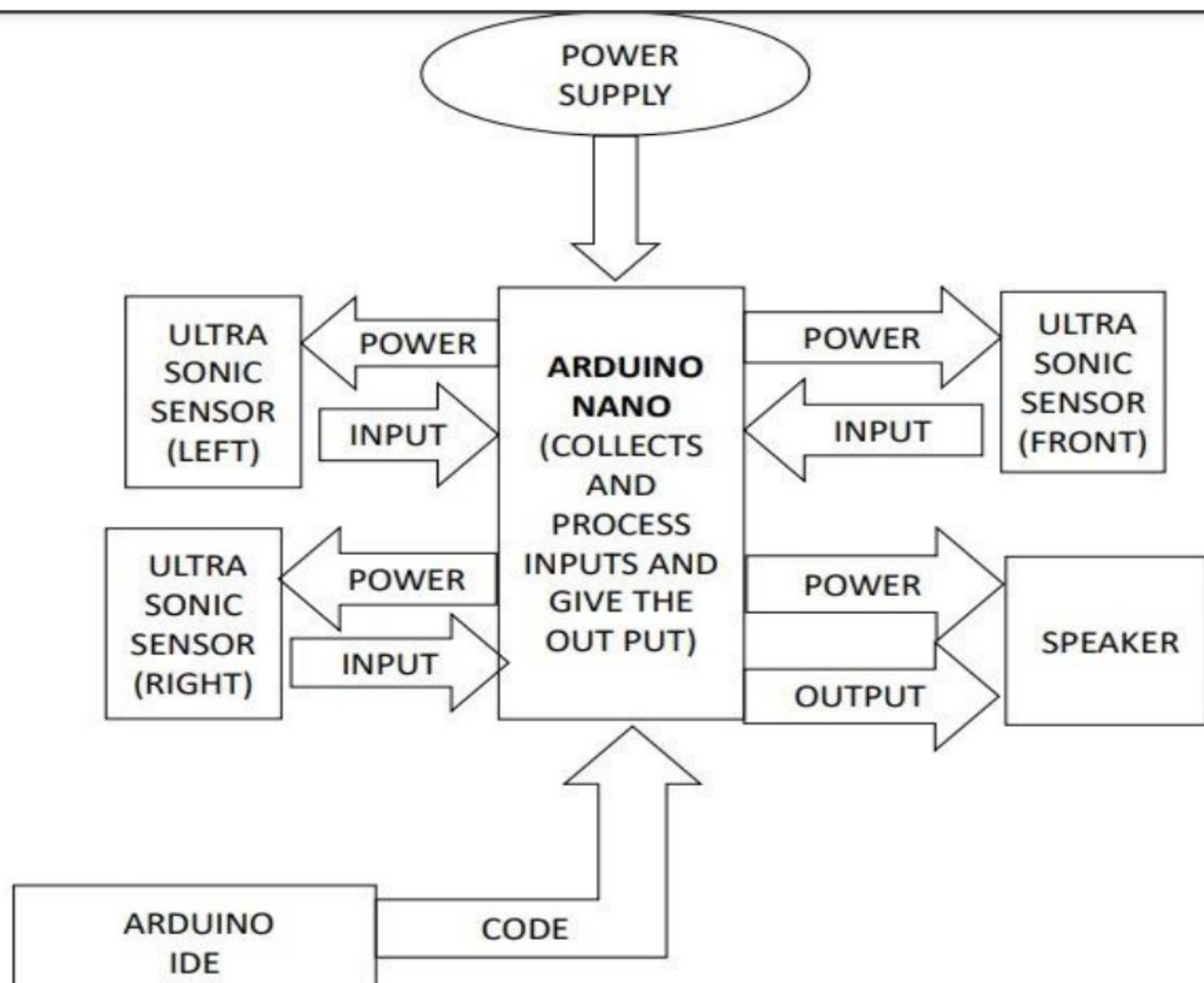


FIG (5) Flowchart.

Chapter -4

Result Analysis and Validations

4.1. Methodology

- The working behind this visually impaired stick is that it is utilized for a specific reason as a detecting gadget for visually impaired individuals. The circuit gives a 5V force supply to the circuit and keeps up with its yield of the force supply at a steady level. It is utilized broadly to recognize objects utilizing ultrasonic sensors. If any object is present, the ultrasonic sensor detects the object by measuring the distance between the object and the user and sends the data to the Arduino Nano.
- To decide the distance of an item, compute the distance between conveying the message and getting back the sign i.e. $\text{Distance} = \text{speed} \times \text{time}$. The speed of the sign going through air is 341m/s. The time is determined between conveying and getting back the message. Since the distance travel by the signal is double, it is divided by two i.e. $\text{Distance} = \text{Distance} / 2$.
- It is placed at the right, left and front of the stick to detect the object. Since, it is especially little reached, it perceives the closer fights. Arduino measures with this information and computes with the order conditions. If any object is found nearer, it sends the command to the user through the speaker or microphone. The order is as of now put away in the voice playback module which sends a ready message to the client about the article.

- **End result and output**

The end result of this project is a user friendly blind stick that helps the user to walk freely in the crowd independently and with a new confidence in them. The technology which helps the user to detect the obstacles within the range of 12 inches and 180 degree sensing technology in a very cost efficient manner will surely help those who are facing different challenges in their life due to blindness.



FIG (5) Blind Stick with Ultra Sonic Sensors.

Chapter -5

Conclusion and Future Work

- **5.1. CONCLUSION**

In the end of our project, we can conclude that our project can reduce the number of risk and injuries for the visually impaired person when walking at public. Nowadays, even at young age experience the visually impairment. This thing cannot be taken so lightly as they know how much risk could it be. If the number of risk and injuries increasing rapidly, the kid or the person will loss their spirit to walk independently. The Modern Blind Stick acts as a basic platform for the coming generation of more aiding devices to help the visually impaired to navigate safely both indoor and outdoor. It is effective and affordable. It leads to good results in detecting the obstacles on the path of the user in a range of two meters. Though the system is hard-wired with sensors and other components, it's light in Conclusions meters. Though the system is hard-wired with sensors and other components, it's light in weight. Further aspects of this system can be improved via wireless connectivity between the system components, thus, increasing the range of the ultrasonic sensor and implementing a technology for determining the speed of approaching obstacles.

- **5.2. FUTURE SCOPE**

The most promising topics that could be the subject of extended analysis are the following research points:

- Further enhancements to boost system performance will be made in future.

- These include: an international system for locating the individual via the GPS and GSM system in order to reach the parent or caregiver venue. It should be flexible and wide range of handling.

REFERENCES

- [1.] Johann B., Iwan U., —The Guide Cane — A Computerized Travel Aid for the Active Guidance of Blind Pedestrians|| proceedings of the IEEE International Conference on Robotics and Automation, Albuquerque, NM, 1283-1288, 1997
- [2.] Koley, S. & Mishra, R., 2012. VOICE OPERATED OUTDOOR NAVIGATION SYSTEM FOR VISUALLY IMPAIRED PERSONS. International Journal of Engineering Trends and Technology.
- [3.] Osama Bader AL-Barrm International Journal of Latest Trends in Engineering and Technology(IJLTETJ)
- [4.] Smart Cane Assisted Mobility for the Visually Impaired, World Academy of Science, Engineering and technology International Journal and Information Engineering Vol:6 No:10,2012
- [5.] G. Prasanthi and P. Tejaswidho Sensor assisted Stick for the Blind People-Transmission on Engineering and Sciences, vol:3,number 1, pp-, 12-16,2015.
- [6.] Ulrich, J. Borenstein, The GuideCane –Applying Mobile Robot Technologies to Assist the visually impaired, IEEE Tr.SMC, Vol:31, No.2, March 2001.
- [7.] Designing an ultrasonic stick prototype for blind people -SULARSO BUDILAKSONA(UNIVERSITY OF PERSADA , INDONESIA
- [8.] Smart Ultrasonic Walking Stick for Visually Impaired People -Manisha Gend(Zeal College of Engineering and Research, Pune)
- [9.] Review on smart blind stick -Aravind Lal
- [10.] Smart blind stick design and implementation -Amira.A.Elsonbaty
- [11.] Walking aid stick for visually challenged people -Vinutha H

[12.] Smart Blind Stick Using Ultrasonic Sensor -T. Tirupal

[13.] Smart Blind Stick For Obstacle Detection and Navigation System -Amit Kumar Thakur